

optcon Semantic Contract & Verification Architecture

Application & Optimization
(Laser Cavity, Inverse Design, Optical RL)



Closed-Form Physical Optics Toolkit
(Gaussian, Cavity, Polarization, Diffraction, Thin-Film, Fiber, MTF)



Type & Contract Enforcement Layer
- Angle as Base Dimension - Amplitude Order Monoid (0, 1, 2)
- Physical Invariants (Unitary, Passive, Reciprocal) - Claerbout Adjoint Check



External Solver Adapter & Differential Testing Registry
(miepython, PyMieScatt, LightPipes, tmm_core, tmm_fast, optiland, poppy, ceviche)